

Final Design and Production of a 10 T HTS Energy-Saving Dipole Magnet for the Italian Facility IRIS

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Abstract—The Innovative Research Infrastructure on Applied Superconductivity (IRIS) is a project funded by the Italian Ministry for University and Research, led by INFN, and coordinated by the LASA laboratory. This project, currently in its final phase, aims at advancing superconductivity for both fundamental research and societal applications, particularly in medical physics and energy. Within this framework, IRIS includes the design and construction of an energy saving, fully high-temperature superconducting dipole magnet (ESMA). ESMA is intended to be both a technological demonstrator for sustainable, cryogen-free magnets, and a user magnet for the IRIS laboratory at Genoa, Italy. This article describes the final design of the dipole and its fabrication status in ASG Superconductors S.p.A. The electromagnetic design consists of 12 flat racetracks, and the target central field at operation is of 10 T. To withstand Lorentz forces in operation, a mechanical structure made by high-strength alloys has been designed and produced. The design of the cryogen-free cooling system is also proposed. Finally, the protection scheme is discussed, together with the incoming tests and operation.

Index Terms—High-temperature superconductors, HTS magnets, dipole magnet, cryogen-free cooling, IRIS project, magnet design, quench protection.

I. INTRODUCTION

THE Italian Ministry for University and Research launched, in October 2022, a major initiative to establish an *Innovative Research Infrastructure on applied Superconductivity* (IRIS) [1], [2] funded under the NextGenerationEU framework. This national infrastructure brings together laboratories from

three major Italian research institutions and five universities, coordinated by INFN (Istituto Nazionale di Fisica Nucleare) through LASA Laboratory (Laboratorio Acceleratori e Superconduttività Applicata) in Milan. The project will finish in April 2026, thanks to a recent extension of the Italian NextGenerationEU projects, providing additional 6 months with respect to the initial timeframe of 3 years.

IRIS aims at both strengthening Italian contribution to future international research programs such as the Future Circular Collider [3] (FCC) at CERN and the Muon Collider [4], and promoting the development of superconducting technologies for broader societal applications, particularly in the energy and medical sectors [5].

Among the key deliverables of the IRIS project there are two major devices: (i) A 130-meter-long Green Superconducting Line, designed to operate at 40 kA and 25 kV, enabling up to 1 GW power transfer for efficient energy applications [6]; (ii) a 10 T HTS dipole magnet for a cable test facility. It also incorporates several design features relevant to future particle accelerators, but intended to operate at 20 K instead of 1.9 K, thereby significantly reducing operational energy consumption [7]. This magnet thus serves as both a technological demonstrator for advancing HTS applications in accelerator magnets (both high-field and medical applications) and as a background field source for the laboratory at INFN Genoa [2]. This paper presents the final design and production status of the 10 T dipole magnet.

II. CONCEPTUAL AND ELECTROMAGNETIC DESIGN

For preliminary design, two key constraints were defined: to ensure that the device could be completed within the IRIS project timeline and, for operations at INFN Genoa, to guarantee reasonable charging times (no more than one hour per tesla). Taking also into account the complexity of constructing a 10 T dipole magnet, posing challenges comparable to a 20 T solenoid, the main design choices, as discussed in [7], are:

- The cable is made of a stack of REBCO tapes [8].
- The coils are flat, single-pancake racetracks.
- The winding incorporates partial insulation via metallic banding, often referred to as metal-as-insulation [9].

Within these principles, key parameters (current, number of turns, number and geometry of racetracks) have been optimized

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TABLE I
ESMA TARGET PERFORMANCES

Central field B_0	10 T
Free aperture	70 mm
Good field region uniformity	$\pm 1.5\%$
Good field region	50 mm $\times$ 30 mm $\times$ 350 mm
Nominal operating temperature	20 K
Critical current margin	>20%

TABLE II
ESMA DESIGN PARAMETERS

Current	1150 A
Energy	3.5 MJ
Coil-to-coil aperture	104 mm
Inductance	5.3 H
Radial Resistance	5.5 $m\Omega$
Time constant τ	16 min
Target ramp-rate	50 mA/s
Coil peak T (static/ramp)	12 K / 18 K

to fulfill the requirements from Table I, resulting in the parameters reported in Table II and discussed in the remaining part of this section. With respect to the conceptual design, as in [7], the iron insert was removed and the conductor is now distributed into six coils per side.

The final cable consists of two face-to-face 12-mm YBCO tapes from Faraday Factory Japan, co-wound with 26 μ m-wide commercially-available AISI 304 tape. The target current for this cable is 1150 A, and the mean thickness of the cable is 162 μ m, resulting in a coil current density of 570 A/mm².

Each racetrack consists of 350 turns of cable, corresponding to around 520 m. The inner mandrel radius is 130 mm and the straight part is 240 mm. Consistently with the actual industry status, the procured tape covers unit lengths in between 200 m and 400 m. This requires internal splicing for the cable. An innovative splicing technology has been developed and is currently in the patenting process. This solution achieves zero over- or under-thickness at the splice (the major drawback of bridge joints), while also avoiding the full local loss of a tape layer (the main issue with lap joints, here retaining 40% of second tape) [10].

The magnet consists of two packs of 6 racetracks each, with 9 mm of distance in between them. The electrical connections in between the racetracks are with copper elements. The geometrical peak-to-peak homogeneity in the good field region is of 2.4% and it is expected to remain within the targets even when accounting for magnetization effects and the mechanical tolerances associated with production and assembly.

In fact, for magnetization, persistent current distributions are available from simulations such as the one shown in the next section, while for the mechanical tolerances a statistical approach was adopted. Assuming experience-based deviations for the major coil-positioning tolerances, a Monte Carlo method was used to verify that a realistic coil geometry would satisfy the requirement.

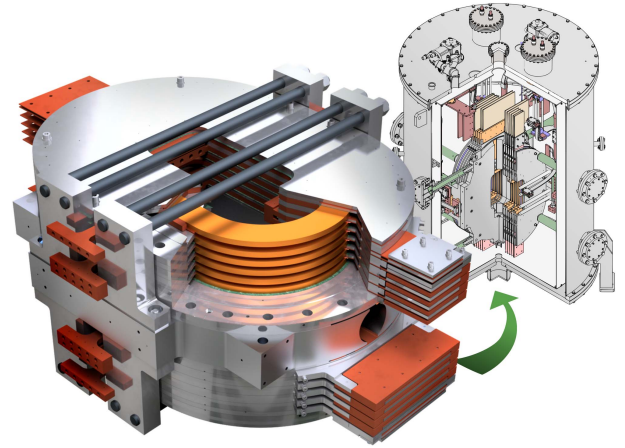


Fig. 1. A render of the cold-mass of ESMA, with a section showing the internal layout of the coils: the upper six-coil pack is partially visible. On the background, cold-mass inside ESMA cryostat. Notice the vertical installation.

III. MECHANICAL, THERMAL, AND PROTECTION DESIGN

The peculiarities of this magnet pose significant challenges across nearly every subsystem:

- The mechanical structure must withstand large Lorentz forces, due to the size, magnetic field, and current density of the device [11].
- The cooling system must work without cryogenic fluids and thus with a very limited cooling budget [12].
- The protection scheme, despite the theoretical advantages of the non-insulated approach, must work on a device of much larger energy than the usual magnets where these principles apply [13].

A. Mechanical Structure and Stress Analysis

The mechanical structure follows the modular approach of the electromagnetic design. Each coil is encased within a support structure primarily composed of AISI 316LN stainless steel. This structure, as shown in the left panel of Fig. 2, pushes presser structures against the coil, providing most of the radial support during cool-down and operation. Axial forces are instead counteracted by the full stack structure (right part of Fig. 2). Detailed structural analyses were performed using 3D finite element models in ANSYS. These simulations evaluated stress distribution, especially under maximum field (forces in Fig. 3). Despite the extensive use of high-strength materials, some components (the ones labeled in bold in Fig. 2) exhibit a safety factor of peak equivalent (von Mises) stress against Ultimate Tensile Strength in between 1.5 and 2. The lowest-safety area is a localized spot on the coil plate, at the edge of the connection with the stacking rods. However, even assuming that this spot plasticizes, the surrounding material has been evaluated to be more than enough to accommodate the additional load due to stress redistribution, at least in a static case. Nevertheless, given the risks coming from this scenario throughout the whole life cycle, this analysis will be updated once preliminary tests (included the prototype

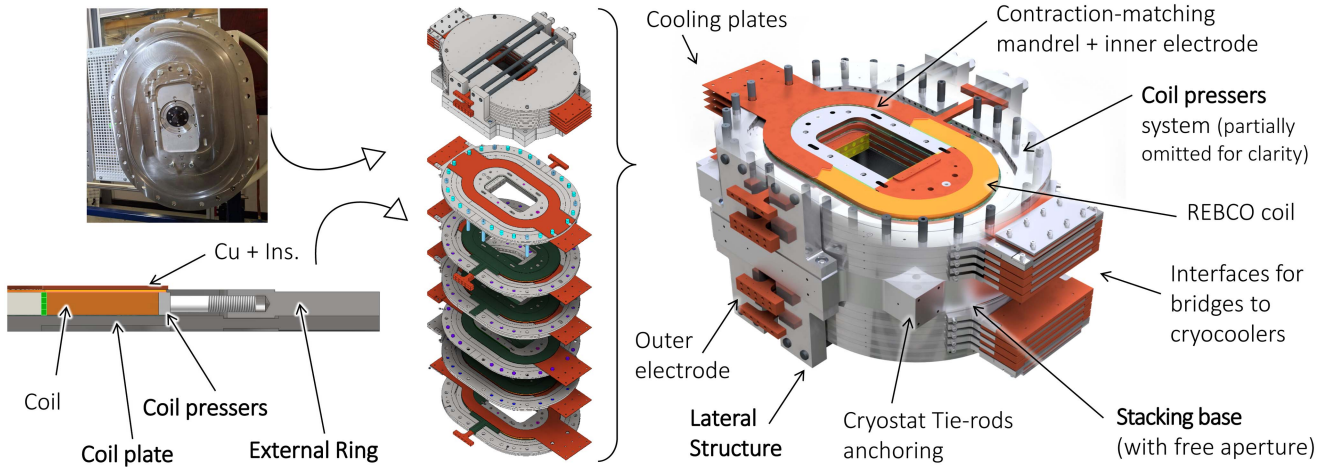


Fig. 2. Overview of the structural design of ESMA. From left to right: the electromagnetic forces containment system for a single coil, the stacking of the modules, and the main components in the final assembly. The elements in bold are further commented in Sec. III-A.

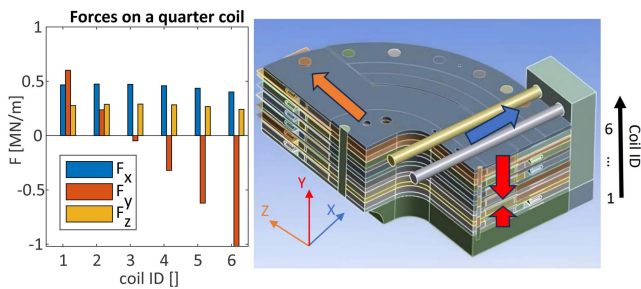


Fig. 3. Overview of the integrated Lorentz forces on a quarter magnet.

module) are made. The lower-than-expected mechanical modulus of the winding pack, of a few GPa [14], is being solved by pre-production winding tests.

B. Thermal Management and Cryogenics

The magnet is entirely conduction-cooled, using three cryocoolers. Thermal bridges linking the coils to the cold heads combine BSCCO current leads and brass/copper components [15]. These elements are pictured in Fig. 4. The main challenge for the operation in cryogen-free mode, i.e. the limited cooling power, is exacerbated in this magnet by the non-insulated winding [16]. In fact, during charging/discharging, the finite contact resistance in between turns results in a radial component of the current generating additional heat. The reference cycle, targeting 15 W, runs the magnet from zero to 10 T in about 6 hours. This has been evaluated with full time-domain 3D models of the magnet from Little Beast Engineering [17]. The model includes the current redistribution within superconductors. A snapshot of this calculation is provided in Fig. 5. To offer a complete discussion of the topic, full simulation results are going to be published together with the outcome of the measurement campaign anticipated in the next section.

Given the importance of the contact resistivity for the magnet, it was investigated experimentally. An hydraulic-press system was employed to estimate the main properties of the winding,

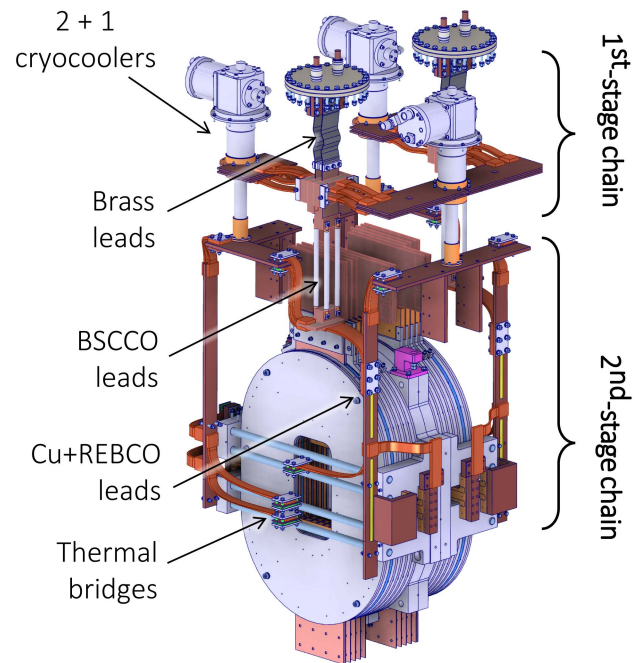


Fig. 4. Overview of the cooling system for ESMA, highlighting the main electrical and thermal elements.

using stacks of ESMA cable (two REBCO and one steel) as specimen, as function of the pressure. From the mechanical analysis, the maximum expected pressure is 150 MPa, at which a specific contact resistance of about $0.24 \text{ m } \Omega \text{ cm}^2$ is found. At the minimum measured pressure of 1 MPa, this value is 100 times lower. Nevertheless, in the pressure region of interest for ESMA in operation (50–150 MPa), contact resistance is weakly dependent on pressure (consistent with the findings in [18]) so $0.24 \text{ m } \Omega \text{ cm}^2$ is assumed for the whole magnet, resulting in the parameters in Table II. It is interesting to highlight that, as in Table II, the τ of the system is much shorter than the 6 hours ramping time. The bottleneck for charging speed is thus not the electromagnetic transient, but the generated heat.

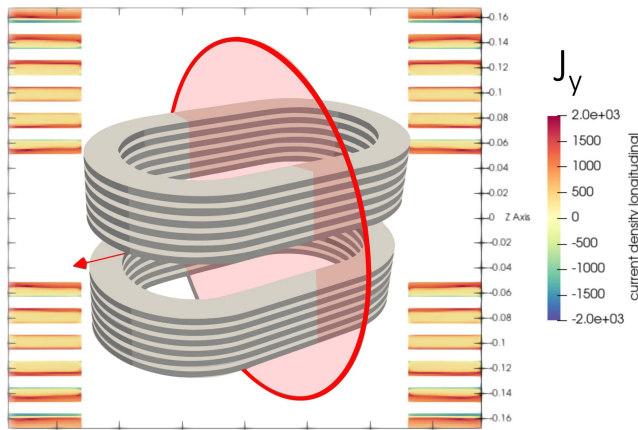


Fig. 5. Current density distribution within cables at the end of the second ramp, as seen in the middle of the straight section.

Simulations for low-amplitude cycles (like 1 T adjustments) are still ongoing, but we expect to fall near the one-hour-per-tesla target even considering the time for field stabilization: shorter ramps are less stressing on the cooling system, and give no time for radial currents to fully develop.

Coil charging losses are by far the major contribution to the losses in cold-mass (second-stage chain in Fig. 4). The second most relevant one, being the Joules heating in copper electrical connections, is about 4 W, with the remaining components (radiation, heat conduction,...) dissipating 1 W. As in Table II, the peak coil temperature is close to the target during ramp, but it considerably drops at steady state, consistently with ramp-up being the major heat source.

Considering auxiliary systems (the first-stage chain in Fig. 4), the major contribution comes from the resistive part of current leads, dissipating 115 W, followed by thermal shield radiation (despite 30 layers of MLI) at about 20 W. All the other contributions do not reach 0.5 W. The BSCCO part of current leads are expected to peak at 56 K, with 6 K margin.

C. Quench Protection Strategy

It is well known that quench in HTS follows a different phenomenology than LTS ones. Minimal quench energies in HTS are in fact 2-3 order of magnitude larger than those in LTS [19]. This makes the magnet more robust to quench, but quench is far more difficult to detect and magnet to protect. Not insulating coil turns is a simple yet effective way to allow the current to bypass a quench hotspot, giving it enough time to be re-absorbed without severe damages. This aspect motivated the metal-as-insulation approach in the preliminary design. Nevertheless, deeper studies highlighted some major issues:

- It is very risky to assume full self-protection. With 3.5 MJ of stored energy and some regions of the magnet peaking a +40 K temperature margin, it is impossible to simply have the energy spread uniformly at quench.
- We evaluated, as protection schemes against an artificially induced quench via a heater (with estimated power of 750 W for 50 ms), both power supply voltage limit and the classical detection and discharge chain (up to 10 mV

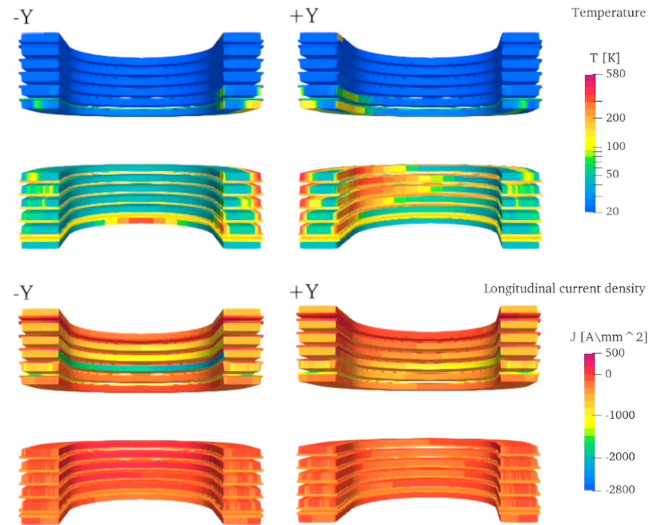


Fig. 6. A snapshot of a Little Beast Engineering simulation showing a quench, triggered in the bottom coil, running in ESMA magnet, showing the coil heating (top) and the current shock-wave propagation (bottom).

threshold within 10 ms delay). While non-insulation allows to mitigate thermal effects, no scenario was safe from a disruptive mechanical shockwave (Fig. 6).

- The only conclusion compatible with IRIS plans and times was a paradigm shift.

The solution for ESMA protection is therefore to switch from an *active* approach to a *preventive* one. An online system, supported by field and temperature sensors, estimates the real-time heat dissipation of the coils. The maximum allowed is the already mentioned 15 W. If the estimation goes above this value, the system intervenes, bypassing the operator to lower the ramp-rate. If this is still not enough, the system is allowed to stop the charging. Finally, in case of an emergency stop, probably the worst-case scenario, the maximum safe ramp-rate (at least a few A/s) is currently under estimation, to allow for a fast magnet switch off without damages. The development of this system strongly relies on industrial know-how of ASG Superconductors S.p.A., and intellectual property management is still under discussion. Once settled this aspect, together with advancements in INFN general research line on real-time magnet controls [20], [21], we will better share our results on the topic with the community.

IV. PRODUCTION, TEST SCHEDULING AND OUTLOOK

As of mid-2025, all mechanical, electrical, and thermal sub-systems, and the winding tooling, are in place at ASG. The cryostat and thermal shield (Fig. 1) are expected before the end of 2025. The refinement of winding procedures is currently ongoing on full-scale mandrels, to optimize the winding process based on the actual properties of the tape, i.e. the lower-than-expected Young's modulus of the stack and the true cross-sectional geometry of the conductor.

Before magnet production, a prototype module is fabricated in the third quarter of 2025. This module will be tested at LASA in a liquid helium environment [22] as soon as it is ready. The goal

is to push the current to the mechanical limit of the coil. Peak stress on the conductor is estimated to occur at 2200 A; once reached, higher currents will be tested to stress the mechanical structure and evaluate its performances.

The final test of the full magnet, within its operational cryostat, will also be conducted at LASA. This will include: cool-down and warm-up cycles, Ramping sequences and stability checks, and magnetic field mapping using a fixed Hall probe array. These tests will verify performance against design predictions and explore the magnet operational limits. The 10 T HTS dipole magnet being developed under the IRIS project represents a first-of-its-kind work toward sustainable high-field magnet technology. It embodies many key innovations: cryogen-free operation, all-HTS conductors, and an innovative approach to quench protection. Despite tight scheduling imposed by IRIS timeline, milestones are generally on track. The magnet is expected to be assembled by the end of 2025, to then proceed with the final tests and commissioning. In addition to being an asset for INFN Genoa, ESMA operation will feed the research activities toward magnets for physics, societal, and medical applications.

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